

Eigenstate based homogenization

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ABSTRACT

The goal of this work is to develop a homogenization approach that achieves the computational efficiency of unresolved (reduced-order) schemes while retaining the accuracy of fully resolved computational homogenization. In the proposed Eigenstate-Based Homogenization (EBH), the fine-scale solution space is represented in terms of eigenstates—such as plastic multipliers or damage parameters—resulting in far fewer unknowns than in finite element discretizations. The representative volume element (RVE) response is expressed as a linear combination of pre-computed eigenmodes, together with partition-level state variables updated to satisfy local consistency, enabling accurate recovery of both fine- and coarse-scale fields.

The nonlinear evolution of the state variables is solved using a Controllable Fixed-Point (CFP) iteration. For plasticity, the spectral radius of the iteration matrix scales linearly with the plastic-multiplier increment; for regularized damage, convergence depends on the load increment, maximum damage, and viscous regularization, ranging from sublinear to superlinear. The resulting tangent operator of the nonlinear system of equation of eigenstates has a condition number several orders of magnitude smaller than that of finite-element direct homogenization, yielding substantial computational savings. To further reduce computational cost, we study a truncated variant of the fixed-point method, termed the Truncated Fixed-Point (TFP) scheme, which leverages the rapid initial error decrease of CFP to obtain low-tolerance solutions with far fewer iterations. Its CPU time is comparable to well-known reduced-order homogenization approaches employing one partition per phase with calibrated properties to fit coarse-scale behavior, yet its accuracy approaches that of fully converged CFP solutions.

Numerical studies show that EBH reproduces both fine-scale fields and macroscopic responses in close agreement with fully resolved finite-element simulations, but at a fraction of the computational cost.

1. Introduction

Homogenization methods can broadly be classified into two categories: resolved and unresolved fine-scale homogenization methods, often referred in the literature to as direct and reduced-order homogenization, respectively. To distinguish these from non-homogenization multiscale approaches—such as domain decomposition or multigrid-like methods—we clarify their defining assumptions and distinctions below.

At the core of all homogenization methods lies the scale separation assumption, which posits that a problem with oscillatory spatial coefficients can be decomposed into a coarse-scale (macroscale) problem and a fine-scale (microscale) or representative volume

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element (RVE) problem. Formally, if l_f and l_c denote the characteristic fine- and coarse-scale lengths, respectively, homogenization assumes $l_f/l_c \ll 1$. In practical settings, such as fiber-reinforced composites or polycrystals, this separation is finite yet sufficient.

Resolved homogenization methods explicitly model RVE geometry and phase constitutive laws to capture the coarse- and fine-scale response. In contrast, unresolved homogenization methods idealize RVE behavior, seeking to predict the effective response without resolving fine-scale details.

Substantial progress in resolved methods has been achieved over the past three decades, especially with the introduction of FFT-based homogenization [1], which solves the Lippmann–Schwinger equation [2–5] efficiently using the Fast Fourier Transform [6]. Despite such advances, the computational cost of fully resolved nonlinear analyses remains prohibitive for large-scale applications, even with improved solvers and numerical stabilization techniques for high-contrast materials [7].

Unresolved homogenization methods achieve computational efficiency through idealizations of microstructure geometry and kinematics. Early examples include the Voigt (uniform strain) [8] and Reuss (uniform stress) [9] assumptions, which provided remarkably useful estimates before the computing era and remain relevant for quick property estimation.

A major subclass involves geometric idealizations, notably the ellipsoidal inclusion models, including Eshelby’s equivalent inclusion [10], Mori–Tanaka [11], and self-consistent schemes [12], which leverage closed-form solutions for ellipsoidal inclusions under eigenstrain loading. These methods, comprehensively reviewed by Nemat-Nasser and Hori [13], approximate real microstructures by ellipsoids and yield analytical tractability at modest cost. Other geometric frameworks, such as Aboudi’s Method of Cells and its generalizations [14,15], or Voronoi cell discretizations [16,17], employ kinematic constraints and weak enforcement of equilibrium. As the discretization is refined, these approaches approach the fidelity, but also the cost of resolved homogenization methods.

Another influential family of idealizations, Transformation Field Analysis (TFA), pioneered by Dvorak and collaborators [18,19], extends Eshelby’s concept to nonlinear materials by representing inelastic effects (plasticity, damage) as evolving transformation fields known as eigenstrains or eigenstresses and treated as internal variables. Model reduction is achieved by approximating eigenstrains via a limited number of basis functions satisfying partition of unity. While computationally efficient, coarse eigenmode approximations (e.g., constant per phase) can lead to locking phenomena [20–24]. We refer to [25–28] and [29] for extension to nonuniform eigenmodal functions and for eigenfields describing interface decohesion.

Another class of unresolved homogenization methods is based on wavelets [30,31], which decompose solution space into coarse- and fine-scale fluctuations using wavelet bases without relying strictly on periodicity or infinite separation of scales. However, nonlinear problems introduce coupled challenges of scale entanglement, costly oscillatory integration, and repeated fine-scale updates [32].

AI/ML tools are often used to augment, but not replace the physics-based homogenization methods including, but not limited to: (i) local law surrogates [33,34], i.e., mapping local constitutive equations subjected to thermodynamics constraints; (ii) homogenized operators, i.e., mapping coarse-scale strains into coarse-scale stresses and consistent tangents throughout the load histories (hyperelasticity [35]; elasto-plasticity–damage [36,37]); and (iii) TFA function surrogates aimed at alleviating locking arising from lower-order approximations [38,39]. Similarly to physics-based homogenization, ML/AI surrogate models for homogenized mapping make the following training-stage idealizations [34]: (i) finite sampling of the state space and load histories, so the learned map mainly interpolates and may fail outside the training hull; (ii) microstructure compression that filters out high-frequency spatial details; (iii) finite-memory history models (small internal state or RNN hidden state) that approximate, rather than reproduce, full path dependence due to information loss; (iv) model-class truncation (low-rank/ROM bases or finite spectral/NN capacity) and associated smoothness bias that can smear yield/damage onsets; and (v) soft physics enforcement where conservation laws are approximated rather than strictly satisfied.

There are several commercial finite-element–based multiscale codes employing various unresolved homogenization techniques, which either idealize microstructural geometry or introduce an approximation of eigenfields. Digimat [40] is based on mean-field homogenization (MFH), which relies on Eshelby’s solution and assumes that the strain in an ellipsoidal inclusion embedded in an infinite homogeneous medium is uniform. It therefore idealizes real microstructures by ellipsoids: fibers approximated as very elongated ellipsoids and platelets as flat ellipsoids. Altair’s Multiscale Designer [41] is based on TFA where an eigenstrain is assumed to be constant over each phase. NASMAT [42] is based on Aboudi’s Method of Cells or its generalizations. Commercial homogenization solvers are mostly physics-based but provide AI-augmented workflows, such as AI/ML-accelerated data generation, inverse identification, and reduced-order surrogates.

We refer to [43–46] for recent review articles on multiscale methods with and without model reduction, the predictive multiscale paradigm for computational design certification, and a posteriori error estimation in multiscale methods.

Despite the broad variety of existing approaches, most unresolved homogenization methods preserve the constitutive framework of the underlying micro-constituents (or phases) but modify their material parameters to match observed macroscopic behavior. This parameter calibration—often referred to as an inverse identification problem—is commonly justified by suggesting that in-situ material properties differ from the nominal, as-manufactured ones. While such variability indeed exists, the discrepancies obtained from calibration frequently arise not from actual material differences but from the simplifying assumptions and idealizations inherent in the modeling framework. Because the influence of these approximations on the overall material response can vary with loading conditions, attributing them to changes in material properties—a practice still widespread in engineering analyses—can be misleading and should be avoided.

The goal of the present work is therefore to develop a homogenization approach that combines the efficiency of unresolved methods with the accuracy of resolved ones. The proposed Eigenstate-Based Homogenization (EBH) method belongs to the TFA family

but introduces two key innovations:

- 1) Representation of fine-scale solution space in terms of eigenstates.

The fine-scale solution in the RVE is expressed as a linear combination of eigenmodes, which are precomputed in the offline stage to satisfy discrete equilibrium equations, and *state variables* (e.g., plastic multipliers, damage parameters, or both). The state variables are discretized and enforced to satisfy consistency conditions.

- 2) Controllable Fixed-Point (CFP) iteration solver.

The nonlinear system of equations for state variables is solved iteratively using a CFP scheme. For plasticity, the spectral radius of the iteration matrix scales linearly with the increment of the plastic multiplier; for regularized damage, convergence rates depend on load parameter, maximum damage, and viscous regularization, ranging from sublinear to superlinear. The condition number of the resulting tangent operator is several orders of magnitude lower than that of conventional finite-element-based direct homogenization, enabling high efficiency without significant loss of accuracy.

As an illustration, Fig. 1 shows the convergence of the residual using the CFP scheme for various plasticity (left) and damage (studied in this work) problems. For damage, the error decreases by roughly an order of magnitude with each iteration, whereas for plasticity it initially drops sharply before slowing and plateauing. In addition to the fully converged CFP scheme, we also examine the truncated fixed-point (TFP) method, which takes advantage of the initial rapid convergence in particular for plasticity problems (Fig. 1a).

The manuscript is organized as follows. Section 2 reviews the fundamentals of two-scale homogenization in the context of plasticity and regularized-damage constitutive models of the fine-scale phases. Section 3 presents the formulation and the overall workflow of the Eigenstate-Based Homogenization (EBH) method, including the transformation from nodal displacements to eigenstate degrees of freedom, and details the Controllable Fixed-Point (CFP) iteration scheme together with its truncated variant (TFP). Section 4 examines the accuracy and computational efficiency of the proposed EBH method relative to the classical FEM-based direct homogenization approach for both plasticity and damage problems using single-fiber and four-fiber RVEs. Section 5 concludes with a summary and directions for future research.

2. Preliminaries

Herein we summarize the results of mathematical homogenization for periodic heterogeneous media in the context of small deformations, and constitutive equations of phases based on plasticity and regularized damage. For model reduction based on transformation field analysis (TFA), the reader is referred to [24].

2.1. Scale separation and periodic RVE formulations

Consider a heterogeneous domain Ω^ε of arbitrary shape subjected to mechanical loading as illustrated in Fig. 2. Due to the underlying microstructural periodicity, the resulting response fields exhibit a characteristic dual-scale behavior: a slowly varying component over the coarse-scale domain, expressed in the coarse-scale coordinate system $\mathbf{x}$, and a rapidly oscillating, periodic response within the representative volume element (RVE), expressed in the fine-scale coordinate system $\mathbf{y}$. The two coordinate systems are related by $\mathbf{y} = \mathbf{x}/\xi$ with $0 < \xi \ll 1$. By performing an asymptotic expansion of the response fields under the assumption of periodicity, one obtains the governing equations for both the homogenized coarse-scale domain Ω and the fine-scale RVE domain Θ .

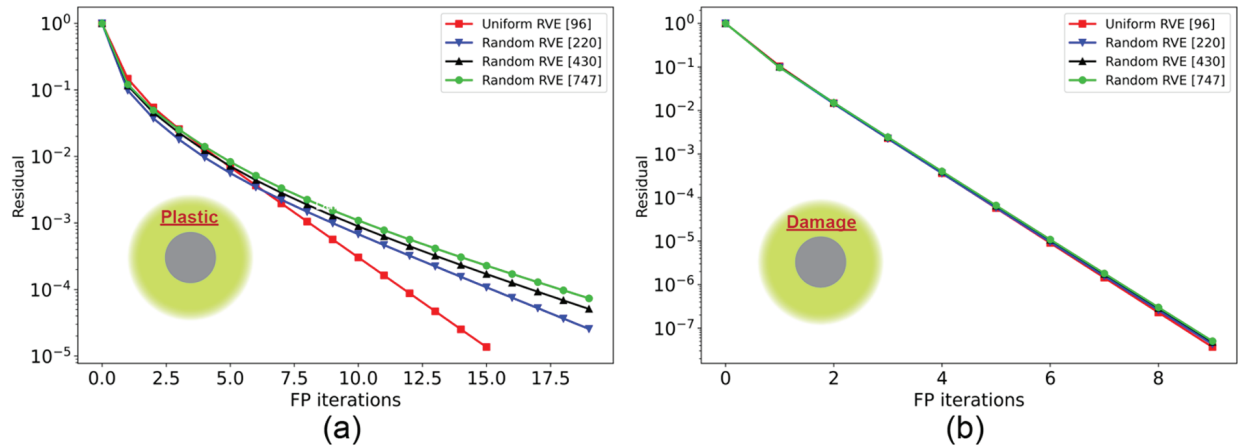


Fig. 1. Convergence of the residual during CFP iterative process. (a) and (b) are for matrix modeled with elastoplastic or damage constitutive laws, respectively.

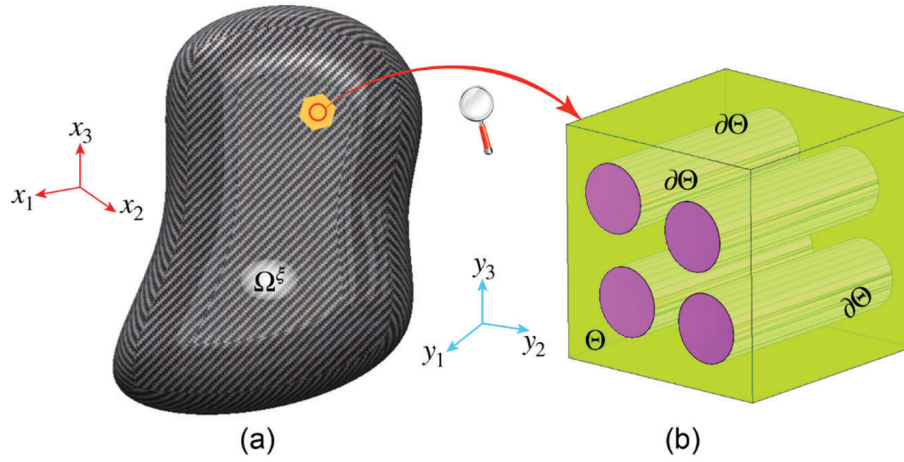


Fig. 2. Schematics of the two-scale problem. (a) composite domain Ω^ε (b) periodic microstructure in RVE domain Θ depicting continuous fiber reinforcement.

Throughout this manuscript, coarse-scale quantities are denoted by an overbar (e.g., $\bar{\varepsilon}$), while without overbar correspond to fine-scale variables.

The strong form of the fine-scale equilibrium equation and its associated boundary conditions are summarized below:

$$\sigma_{ij,j}(\mathbf{x}, \mathbf{y}) = 0 \quad \text{in} \quad \Theta \quad (1)$$

$$\sigma_{ij}(\mathbf{x}, \mathbf{y}) = L_{ijkl}(\mathbf{y})(\varepsilon_{kl}(\mathbf{x}, \mathbf{y}) - \mu_{kl}(\mathbf{x}, \mathbf{y})) \quad \text{in} \quad \Theta \quad (2)$$

$$\varepsilon_{kl}(\mathbf{x}, \mathbf{y}) = \bar{\varepsilon}_{kl}(\mathbf{x}) + u_{(k,y_l)}(\mathbf{x}, \mathbf{y}) \quad \text{in} \quad \Theta \quad (3)$$

$$u_i(\mathbf{x}, \mathbf{y}) = u_i(\mathbf{x}, \mathbf{y} + \mathbf{l}) \quad \text{on} \quad \partial\Theta \quad (4)$$

$$\langle u_i(\mathbf{x}, \mathbf{y}) \rangle = 0 \quad \text{in} \quad \Theta \quad (5)$$

where $\mathbf{u}$ denotes the displacement field, ε the fine-scale strain, σ the fine-scale stress, $\mathbf{L}$ the elastic constitutive tensor of phases, and μ the eigenstrain resulting from either plasticity or damage. The Einstein summation convention over repeated indices is employed throughout.

2.2. Transformation field analysis and discretization of eigenfields

The salient feature of TFA is that the fine-scale displacement $u_i(\mathbf{x}, \mathbf{y})$ is formulated to automatically satisfy the fine-scale equilibrium equation in the RVE for arbitrary coarse-scale strain $\bar{\varepsilon}_{kl}(\mathbf{x})$ and eigenstrain $\mu_{ij}(\mathbf{x}, \mathbf{y})$

$$u_i(\mathbf{x}, \mathbf{y}) = H_i^{kl}(\mathbf{y})\bar{\varepsilon}_{kl}(\mathbf{x}) + \int_{\Theta} h_i^{kl}(\mathbf{y}, \tilde{\mathbf{y}})\mu_{kl}(\mathbf{x}, \tilde{\mathbf{y}})d\tilde{\Theta} \quad (6)$$

where H_i^{kl} and h_i^{kl} are the transformation influence functions chosen to satisfy the strong form of the boundary value problem in the RVE subjected to the coarse-scale strain and fine-scale eigenstrain, respectively.

Model reduction is achieved by eigenstrain μ_{ij} discretization

$$\mu_{ij}(\mathbf{x}, \mathbf{y}) = \sum_{A=1}^M N^A(\mathbf{y})\mu_{ij}^A(\mathbf{x}) \quad (7)$$

where M is the number of modes or partitions. It is customary though not necessary to employ a piece-wise constant discretization, i.e., $N^A(\mathbf{y}) \in C^{-1}(\Theta)$ satisfying the partition of unity condition $\sum_{A=1}^M N^A(\mathbf{y}) = 1$ with μ_{ij}^A denoting constant eigenstrain on partition Θ^A . Inserting Eq. (7) into Eq. (6), taking symmetric spatial derivative and averaging over a partition Θ^B yields the partition level strain

$$\varepsilon_{ij}^B(\mathbf{x}) = E_{ij}^{klB}\bar{\varepsilon}_{kl}(\mathbf{x}) + P_{ij}^{klBA}\mu_{kl}^A(\mathbf{x}) \quad (8)$$

where

$$E_{ij}^{klB} = I_{ijkl} + \frac{1}{|\Theta^B|} \int_{\Theta^B} H_{(i,y_j)}^{kl}(\mathbf{y}) d\Theta \quad (9)$$

$$P_{ij}^{klBA} = \frac{1}{|\Theta^B|} \int_{\Theta^B} \left(\int_{\Theta} h_i^{kl}(\mathbf{y}, \tilde{\mathbf{y}}) N^A(\tilde{\mathbf{y}}) d\tilde{\Theta} \right)_{y_j} d\Theta \quad (10)$$

The Einstein summation convention is also employed for modal (partition) values unless otherwise stated. The discrete transformation functions, E_{ij}^{klB} and P_{ij}^{klBA} , are pre-calculated during the offline stage by solving a sequence of elastic periodic RVE problems, detailed in [24]. Hereafter we omit the dependence on fine- and coarse-scale coordinates.

2.3. Constitutive models of phases

Herein, the constitutive behavior of the fibers is modeled using isotropic damage mechanics, whereas the matrix is represented by either isotropic plasticity or damage.

2.3.1. Isotropic plasticity model

For demonstration purposes, the von Mises plasticity model with linear isotropic hardening is adopted in this work. The yield function is defined as

$$f = \frac{3}{2} \sigma_{ij} \mathbb{I}_{ijkl}^{dev} \sigma_{kl} - (Y_0 + H\lambda)^2 \quad (11)$$

where $\mathbb{I}_{ijkl}^{dev} = \frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl}$ denotes the deviatoric operator, Y_0 the initial yield stress, H the plastic modulus, and λ the plastic multiplier. The evolution of λ is governed by the Kuhn-Tucker conditions

$$f \leq 0, \dot{\lambda} \geq 0, \dot{\lambda} f = 0 \quad (12)$$

which ensures consistency of the plastic state. The corresponding increment of plastic strain $\Delta\mu_{ij}$ is given by

$$\Delta\mu_{ij} = \Delta\lambda \mathbb{I}_{ijkl} \sigma_{kl} \quad (13)$$

where σ_{kl} is an updated stress.

2.3.2. Regularized isotropic damage model

In isotropic damage model, a scalar damage variable $0 \leq \omega < 1$ characterizes the extent of material degradation and governs the reduction of stiffness, thereby inducing a nonlinear stress-strain response. In this work, a strain-based formulation is adopted, in which the loading function is defined as

$$f = \varepsilon_{eq} - \kappa \quad (14)$$

where $\varepsilon_{eq} = \sqrt{\varepsilon_{ij} \varepsilon_{ij}}$ is the equivalent strain - a scalar measure of the elastic strain - and κ is a history-dependent variable recording the maximum value of ε_{eq} attained during the loading history. The evolution of κ satisfies the standard Kuhn-Tucker loading-unloading conditions:

$$f \leq 0, \dot{\kappa} \geq 0, \dot{\kappa} f = 0 \quad (15)$$

The damage variable ω is defined as a function of κ

$$\omega(\kappa) = \begin{cases} 0 & \kappa < \varepsilon_0 \\ \omega_m \left[1 - \frac{\varepsilon_0}{\kappa} \exp\left(-\frac{\kappa - \varepsilon_0}{\varepsilon_f - \varepsilon_0}\right) \right] & \kappa \geq \varepsilon_0 \end{cases} \quad (16)$$

where ε_0 , ε_f , ω_m are material constants. To ensure well-posedness and numerical stability, the damage evolution is regularized in pseudo-time along the lines of viscous regularization. Without such a regularization, strain softening induces severe localization of deformation into arbitrarily narrow bands, leading to mesh dependency and non-unique solutions.

Let $0 \leq \gamma \leq 1$ denote the loading parameter representing the fraction of the total applied load, where $\gamma = 1$ corresponds to full load. For a total of n load increments, the increment size is $\Delta\gamma = 1/n$. The viscous-like regularization of an eigenstrain is defined as through regularization parameter τ , and the regularized eigenstrain is defined as

$$\mu_{ij} = \frac{\tau}{\tau + \Delta\gamma} q \mu_{ij} + \frac{\Delta\gamma}{\tau + \Delta\gamma} \omega(\kappa) \varepsilon_{ij} \quad (17)$$

where τ is regularization parameter; $\frac{\tau}{\tau + \Delta\gamma}$ quantifies the contribution of the converged solution at the previous load increment q on the solution of the current increment. The bottom left subscript denotes the load increment. The left-subscript for the current load

increment $q + 1$ is omitted throughout the manuscript.

Finally, the stress is updated as

$$\sigma_{ij} = L_{ijkl}(\epsilon_{kl} - \mu_{kl}) \quad (18)$$

3. Eigenstate-based homogenization (EBH)

Herein we present the formulation of the Eigenstate-Based Homogenization (EBH) method, including the transformation from nodal displacements to eigenstate degrees-of-freedom, the Controllable Fixed-Point (CFP) iteration scheme and its convergence characteristics, as well as its truncated variant, hereafter referred to as TFP.

3.1. EBH formulation for plasticity

This subsection is devoted to EBH formulation for solving plasticity problems where the eigenstate is a plastic multiplier.

3.1.1. Direct EBH approach for plasticity

Based on the partition-level strain ϵ_{ij}^B (Eq. (8)), the partition-level stress is computed as

$$\begin{aligned} \sigma_{ij}^B &= L_{ijst}^B (\epsilon_{st}^B - \mu_{st}^B) = \Sigma_{ij}^{klB} \bar{\epsilon}_{kl} + C_{ij}^{klBA} \mu_{kl}^A \\ \Sigma_{ij}^{klB} &= L_{ijst}^B E_{st}^{klB} \\ C_{ij}^{klBA} &= L_{ijst}^B \left(P_{st}^{klAB} - \delta_{AB} I_{klst} \right) \end{aligned} \quad (19)$$

where Σ_{ij}^{klB} and C_{ij}^{klBA} are the influence functions, directly following from Eqs. (9)-(10).

Throughout the manuscript the underbar on the partition indices denotes *no summation* over the repeated indices.

At each load increment, the *trial partition-level stress* σ_{ij}^{trB} is obtained by accounting for the contribution of the coarse-scale strain increment $\Delta \bar{\epsilon}_{kl}$ and eigenstrain from the previous load increment $q \mu_{kl}^A$

$$\sigma_{ij}^{trB} = {}_q \sigma_{ij}^B + \Sigma_{ij}^{klB} \Delta \bar{\epsilon}_{kl} = \Sigma_{ij}^{klB} \bar{\epsilon}_{kl} + C_{ij}^{klBA} q \mu_{kl}^A \quad (20)$$

Next, the updated partition-level stress σ_{ij}^B is computed by accounting for the contribution of the plastic strain increment $\Delta \mu_{kl}^A = \mathbb{I}_{klst}^{dev} \sigma_{st}^A \Delta \lambda^A$

$$\sigma_{ij}^B = \sigma_{ij}^{trB} + C_{ij}^{klBA} \mathbb{I}_{klst}^{dev} \sigma_{st}^A \Delta \lambda^A \quad (21)$$

Rearranging Eq. (21) yields the expression for the partition-level stress σ_{ij}^A as function of the plastic multiplier increments $\Delta \lambda$ in all partitions

$$\sigma_{ij}^A = \left[B_{ijst}^{BA}(\Delta \lambda) \right]^{-1} \sigma_{st}^{trB} \quad (22)$$

where

$$B_{ijst}^{BA}(\Delta \lambda) = I_{ijst} \delta_{AB} - C_{ij}^{klBA} \mathbb{I}_{klst}^{dev} \Delta \lambda^A \quad (23)$$

Prior to solving for the unknown plastic multiplier increments $\Delta \lambda$ from the partition level consistency condition, the process in each partition is classified as either elastic or inelastic. Initially all the plastic multiplier increments are assumed to vanish $\Delta \lambda^A = 0$. The process in a partition is considered to be elastic, i.e. $\Delta \lambda^A = 0$, if

$$\frac{3}{2} \sigma_{ij}^A(\Delta \lambda) \mathbb{I}_{ijkl}^{dev} \sigma_{kl}^A(\Delta \lambda) - ({}_q Y^A)^2 < 0 \quad (24)$$

Otherwise, the process in a partition is inelastic in which case the increment of plastic multipliers $\Delta \lambda$ in a partition is computed from the *consistency conditions* defined as

$$r^A(\Delta \lambda) = \frac{3}{2} \sigma_{ij}^A(\Delta \lambda) \mathbb{I}_{ijkl}^{dev} \sigma_{kl}^A(\Delta \lambda) - ({}_q Y^A + H^A \Delta \lambda^A)^2 = 0 \quad (25)$$

in combination with Eqs. (22) and (23). Note that the Newton method couples plastic multiplier increments through the coupling term C_{ij}^{klBA} .

3.1.2. Controllable fixed-point (CFP) iterative method

This subsection introduces controllable fixed-point iterative solution procedure for solving a dense nonlinear system of Eqs. (25). The Newton method requires repeated solution of a dense linear system of equations, Eqs. (22) and (23), which becomes

computationally taxing as the number of partitions increases, motivating the development of an alternative approach that avoids solution of dense system of equations.

3.1.2.1. CFP algorithm. The proposed fixed-point iterative algorithm solves the nonlinear system of equations at each load increment by successive iterations (denoted herein by the left superscript) until convergence is reached.

For each load increment, the initial partition-level trial stress ${}^0\sigma_{ij}^{trB}$, accounting for $\Delta\bar{\epsilon}_{kl}$ only, is first computed from Eq. (20). The initial trial stress is then used to evaluate the initial increment of an eigenstate, ${}^0\Delta\lambda^B$, from the consistency condition and subsequently, compute the initial stress ${}^0\sigma_{ij}^B$

$${}^0\sigma_{ij}^B = \left(I_{ijst} + L_{ijkl}^B \mathbb{I}_{klst}^{dev} {}^0\Delta\lambda^B \right)^{-1} {}^0\sigma_{st}^{trB} \quad (26)$$

It is important to note that Eq.(26) unlike Eqs. (22), (23) is carried out partition-by-partition ignoring the coupling between partitions.

The resulting initial eigenstrain increment ${}^0\Delta\mu_{kl}^A = \mathbb{I}_{klst}^{dev} {}^0\sigma_{st}^A$ is then used to initialize the fixed-point iterative process.

During the fixed-point iteration, the trial stress for the $(m+1)$ -th iteration ${}^{m+1}\sigma_{ij}^{trB}$ is updated based on the eigenstrain increment obtained from the previous iteration ${}^m\Delta\mu_{st}^A$

$${}^{m+1}\sigma_{ij}^{trB} = {}^0\sigma_{ij}^{trB} + L_{ijkl}^B P_{kl}^{stBA} {}^m\Delta\mu_{st}^A \quad (27)$$

The trial stress ${}^{m+1}\sigma_{ij}^{trB}$ is then used to update the plastic multiplier increment ${}^{m+1}\Delta\lambda^B$, based on the consistency condition at each partition, the stress ${}^{m+1}\sigma_{ij}^B$, the eigenstrain increment ${}^{m+1}\Delta\mu_{kl}^B$, and the yield surface

$$\begin{aligned} {}^{m+1}\sigma_{ij}^B &= \left(I_{ijst} + L_{ijkl}^B \mathbb{I}_{klst}^{dev} {}^{m+1}\Delta\lambda^B \right)^{-1} {}^{m+1}\sigma_{st}^{trB} \\ {}^{m+1}\Delta\mu_{kl}^B &= \mathbb{I}_{klst}^{dev} {}^{m+1}\sigma_{st}^B {}^{m+1}\Delta\lambda^B \\ {}^{m+1}Y^B &= {}^qY^B + H^B {}^{m+1}\Delta\lambda^B \end{aligned} \quad (28)$$

The iteration proceeds until the convergence criterion $\| {}^{m+1}\Delta\lambda^A - {}^m\Delta\lambda^A \| < TOL$ is reached.

3.1.2.2. CFP convergence estimate. CFP convergence study consists of bounding the error in the partition trial stress from above and below by an error in the partition stress in two subsequent iterations. We start with the expression of the partition trial stress in the iteration $m+1$

$${}^{m+1}\sigma_{ij}^{trB} = {}^0\sigma_{ij}^{trB} + L_{ijkl}^B P_{kl}^{stBA} \mathbb{I}_{stmn}^{dev} {}^m\sigma_{mn}^A {}^m\Delta\lambda^A \quad (29)$$

Similarly to Eq. (29) the converged partition trial stress $\hat{\sigma}_{ij}^{trB}$ is related to the converged partition stress $\hat{\sigma}_{ij}^B$ and the converged partition plastic multiplier increment $\hat{\Delta\lambda}^B$ by

$$\hat{\sigma}_{ij}^{trB} = {}^0\sigma_{ij}^{trB} + L_{ijkl}^B P_{kl}^{stBA} \mathbb{I}_{stmn}^{dev} \hat{\Delta\lambda}^A \hat{\sigma}_{mn}^A \quad (30)$$

Subtracting Eq. (29) from Eq. (30) yields the error in the partition trial stress ${}^{m+1}e_{ij}^{trB}$ at an iteration $m+1$

$${}^{m+1}e_{ij}^{trB} = L_{ijkl}^B P_{kl}^{stBA} \mathbb{I}_{stmn}^{dev} (\hat{\Delta\lambda}^A \hat{\sigma}_{mn}^A - {}^m\Delta\lambda^A {}^m\sigma_{mn}^A) \quad (31)$$

Taking Euclidean norm on Eq.(31) yields

$$\| {}^{m+1}e_{ij}^{trB} \|_2 \leq \| L_{ijkl}^B P_{kl}^{stBA} \mathbb{I}_{stmn}^{dev} \|_2 \| \hat{\Delta\lambda}^A \hat{\sigma}_{mn}^A - {}^m\Delta\lambda^A {}^m\sigma_{mn}^A \|_2 \quad (32)$$

Further bounding the second term on the right-hand side by

$$\| \hat{\Delta\lambda}^A \hat{\sigma}_{mn}^A - {}^m\Delta\lambda^A {}^m\sigma_{mn}^A \|_2 \leq \| \Delta\lambda_{\max}^A {}^m\sigma_{mn}^A \|_2 \quad (33)$$

where $\Delta\lambda_{\max}^A \geq \max(\hat{\Delta\lambda}^A, {}^m\Delta\lambda^A)$.

Further inserting Eq. (33) into Eq. (32) yields the upper bound

$$\| {}^{m+1}e_{ij}^{trB} \|_2 \leq \Delta\lambda_{\max}^A \| L_{ijkl}^B P_{kl}^{stBA} \mathbb{I}_{stmn}^{dev} \|_2 \| {}^m\sigma_{mn}^A \|_2 \quad (34)$$

For lower bound estimation we employ an error reduction property of return-mapping algorithms, which states that the stress error in the Euclidean norm in after the return-mapping correction is strictly smaller than the stress error in the trial state [47]. It is a trivial exercise to show that the distance between any two points in the trial space outside the yield surface is larger than the distance between the same two points projected onto the convex surface. If the two points are selected as ${}^{m+1}\sigma^{tr}$ and $\hat{\sigma}^{tr}$, we get the lower bound

$$\| {}^{m+1}e_{st}^B \|_2 \leq \| {}^{m+1}e_{ij}^{trB} \|_2 \quad (35)$$

Finally, combining Eqs. (35) and (34) gives

$$\| {}^{m+1}e_{ij}^B \|_2 \leq \Delta\lambda_{\max} \| L_{ijkl}^B P_{kl}^{stBA} \|_{dev} \| {}^m e_{mn}^A \|_2 \quad (36)$$

This completes the proof of monotonic error decrease with a rate of convergence governed by the largest partition plastic multiplier increment, which in turn, can be controlled by adjusting the step size.

3.1.2.3. Truncated fixed-point iteration method (TFP). To further reduce computational cost while retaining sufficient accuracy at a fine scale, a truncated variant of the fixed-point iteration algorithm will be explored. This approach takes advantage of an accelerated rate of convergence in the first two iterations as indicated in Fig. 1, to provide convergence to a lower tolerance. The efficiency and accuracy of the truncated variant, herein denoted as TFP, will be quantitatively assessed in the numerical examples presented in Section 4.

3.2. EBH formulation for damage

This subsection is devoted to EBH formulation for solving damage mechanics problems where the eigenstate is a damage parameter.

3.2.1. Direct EBH approach for damage

Following Eq. (8), the partition level strain at every load increment is computed using the prescribed coarse-scale strain increment $\Delta\bar{\epsilon}_{kl}$ and the eigenstrain increment $\Delta\mu_{kl}^A$ (to be determined iteratively) as

$$e_{ij}^B = q e_{ij}^B + E_{ij}^{klB} \Delta\bar{\epsilon}_{kl} + P_{ij}^{klBA} \Delta\mu_{kl}^A \quad (37)$$

where the eigenstrain increment follows from Eq. (17)

$$\Delta\mu_{kl}^A = \mu_{kl}^A - q\mu_{kl}^A = \frac{\Delta\gamma}{\tau + \Delta\gamma} [\omega_{kl}^A - q\mu_{kl}^A] \quad (38)$$

Inserting Eq. (38) into (37) and collecting all known terms on the right hand side yields the resulting the strain e_{kl}^A in term of the unknown damage variables ω in all partitions

$$e_{kl}^A = [B_{ijkl}^{BA}(\omega)]^{-1} e_{ij}^{trB} \quad (39)$$

where

$$B_{ijkl}^{BA}(\omega) = I_{ijkl} \delta_{AB} - \frac{\Delta\gamma}{\tau + \Delta\gamma} P_{ij}^{klBA} \omega_{kl}^A \quad (40)$$

$$e_{ij}^{trB} = q e_{ij}^B + E_{ij}^{klB} \Delta\bar{\epsilon}_{kl} + \frac{\tau}{\tau + \Delta\gamma} P_{ij}^{klBA} q \mu_{kl}^A \quad (41)$$

Prior to solving for the unknown partition damage variables ω from the consistency condition at the partition level, the process in each partition is classified as either elastic or inelastic. The process in a partition is elastic, i.e. $\omega^A = q\omega^A$, if

$$\sqrt{\epsilon_{ij}^A(\omega) \epsilon_{ij}^A(\omega)} - \kappa^A(q\omega^A) \leq 0 \quad (42)$$

Otherwise, the process in a partition is inelastic requiring solution of the nonlinear system of equations

$$r^A(\omega) = \sqrt{\epsilon_{ij}^A(\omega) \epsilon_{ij}^A(\omega)} - \kappa^A(\omega^A) = 0 \quad (43)$$

3.2.2. Controllable fixed-point (CFP) iterative method

3.2.2.1. CFP algorithm. For each load increment, the initial partition level strain ${}^0e_{ij}^B$ is computed by considering only coarse-scale strain increment $\Delta\bar{\epsilon}_{kl}$

$${}^0e_{ij}^B = q e_{ij}^B + E_{ij}^{klB} \Delta\bar{\epsilon}_{kl} \quad (44)$$

This initial strain is used to compute the initial partition damage parameter ${}^0\omega^B$ from the damage evolution law Eq. (16), and the corresponding regularized eigenstrain increment ${}^0\Delta\mu_{kl}^B$ from Eq. (38), which serves as the starting point for the fixed-point iteration.

During the iterative process, the partition strain for the $(m+1)$ -th iteration is updated as

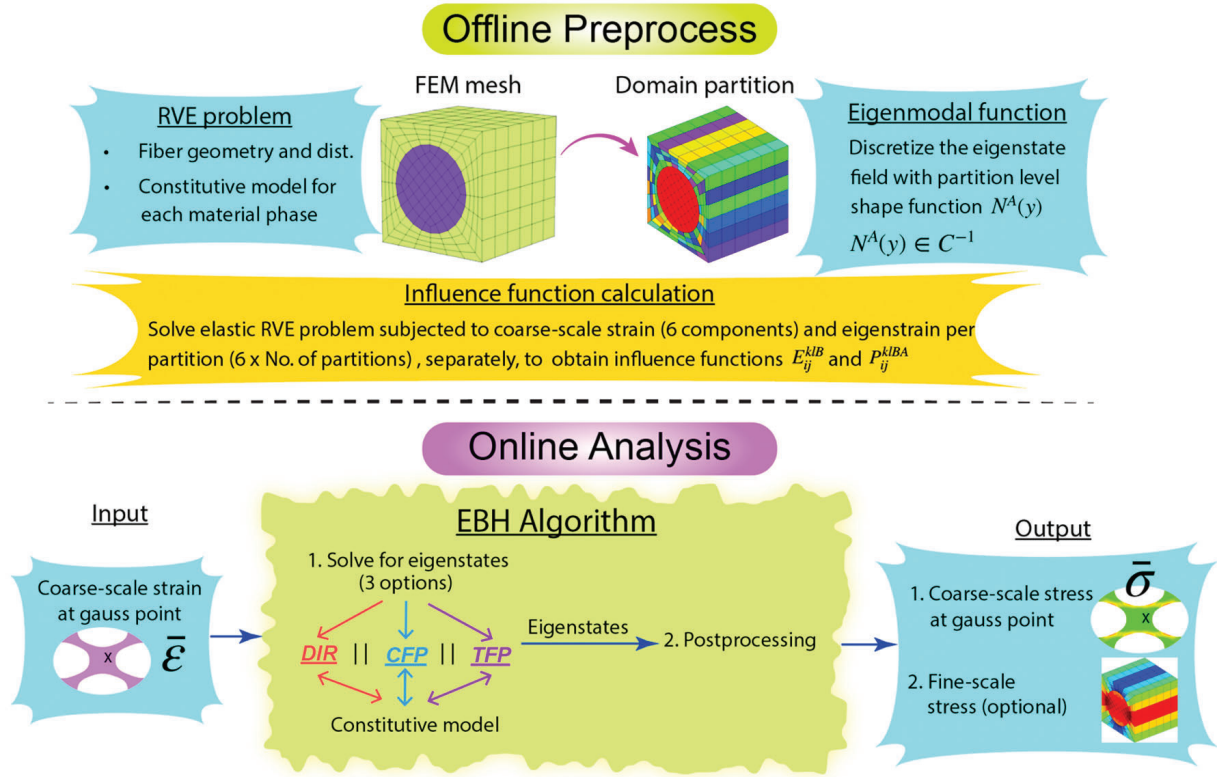


Fig. 3. Workflow of the EBH computational framework. The upper panel depicts the offline preparation stage, aimed at calculating the influence functions for the online analysis stage, depicted in the lower panel.

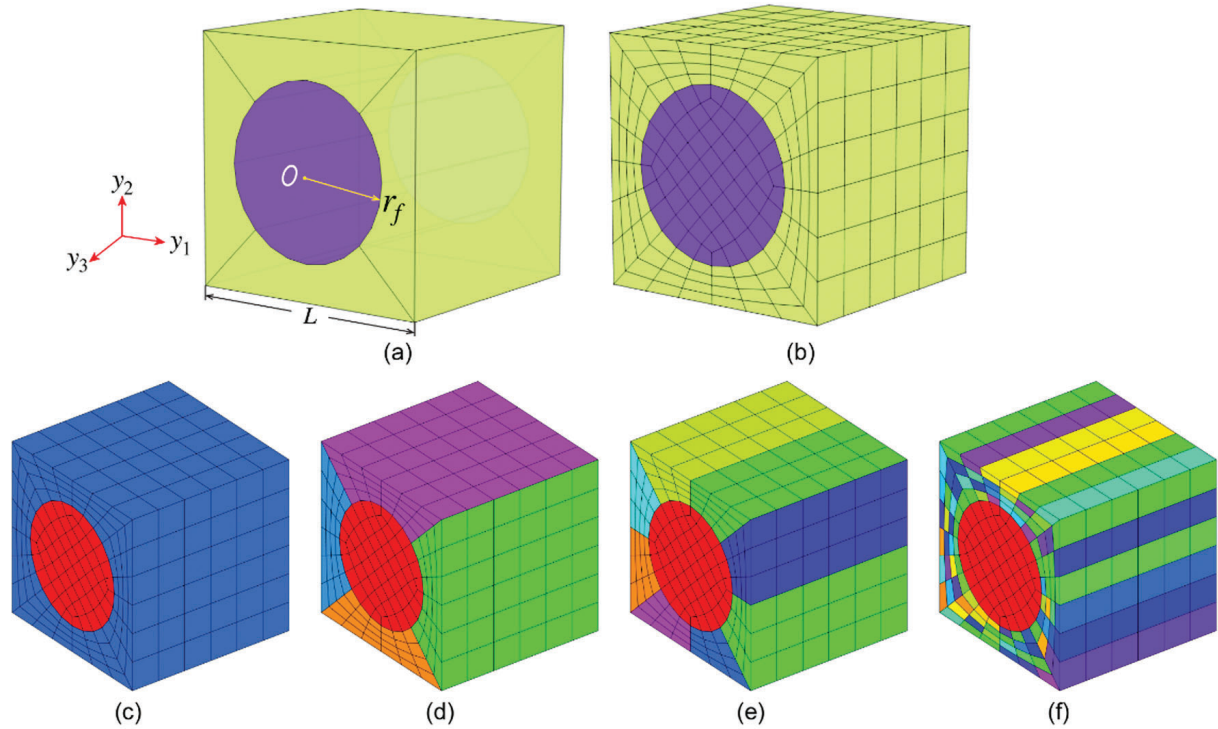


Fig. 4. Single fiber RVE domain and partitioning. (a) Geometry and fine-scale coordinate system of the RVE. (b) Finite element mesh. (c-f) Partitioning of the material phases, where the matrix is divided into 1, 4, 8, and 96 partitions, respectively. Elements marked by the same color belong to one partition.

$${}^{m+1}\varepsilon_{ij}^B = {}^0\varepsilon_{ij}^B + P_{ij}^{klBA} {}^m\Delta\mu_{kl}^A \quad (45)$$

from which the updated damage variable ${}^{m+1}\omega^A$ and the regularized eigenstrain increment ${}^{m+1}\Delta\mu_{kl}^A$ are computed for the next iteration using Eqs. (16) and (38), respectively. The process is repeated until the convergence $\|{}^{m+1}\omega^A - {}^m\omega^A\| < TOL$ is reached.

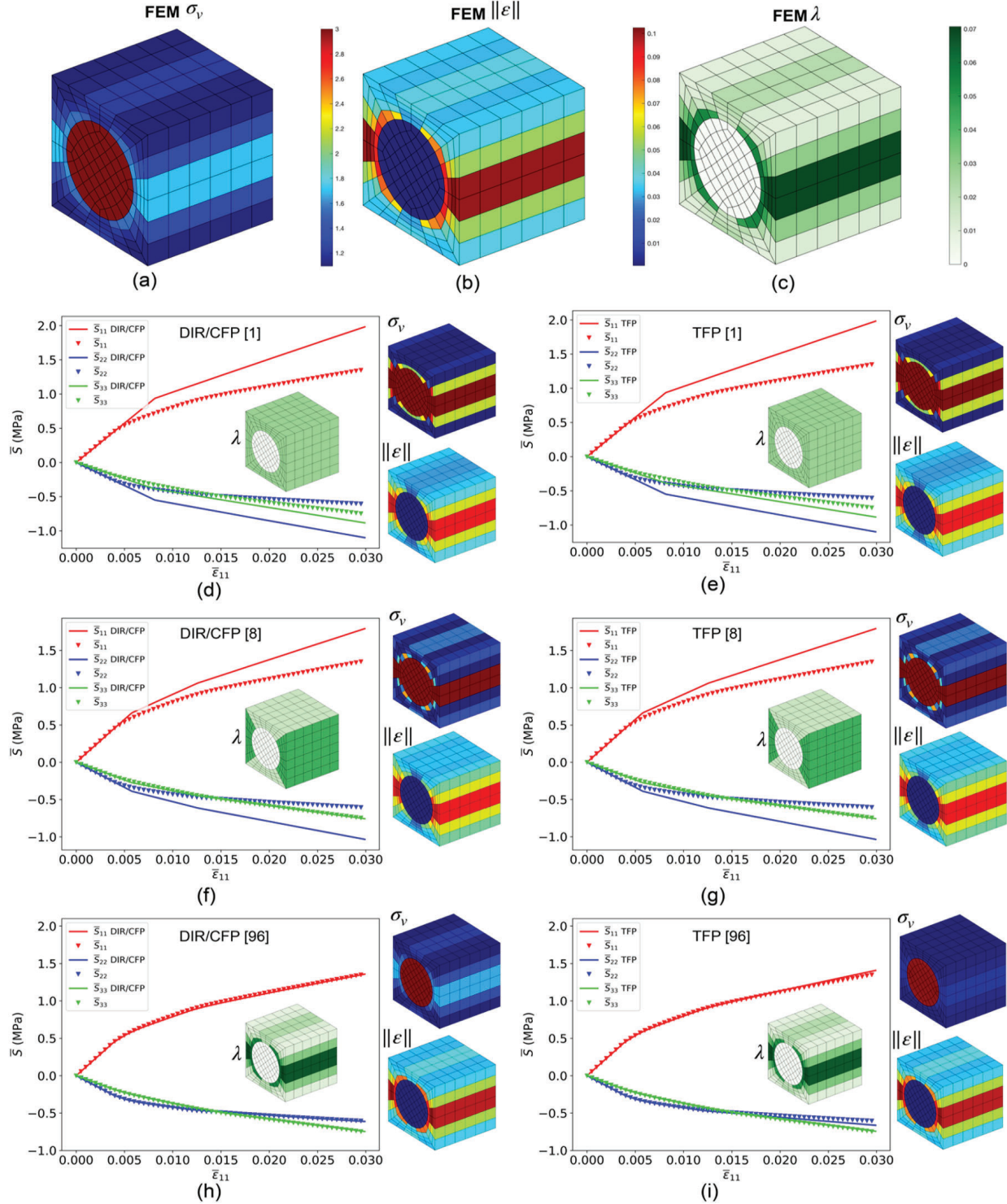


Fig. 5. EBH results for matrix modeled as elastoplastic under uniaxial tension applied normal to the fiber direction. (a)-(c) FEM solutions of the fine-scale von Mises stress σ_v , strain norm $\|\varepsilon\|$, and accumulated plastic multiplier λ , at the end of loading. (d), (f), and (h) depict the DIR/CFP solutions for the coarse-scale stress and corresponding fine-scale fields computed with 1, 8, and 96 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities of interest and partition levels. The color scales of fine-scale fields obtained from EBH correspond to those of the FEM reference solutions.

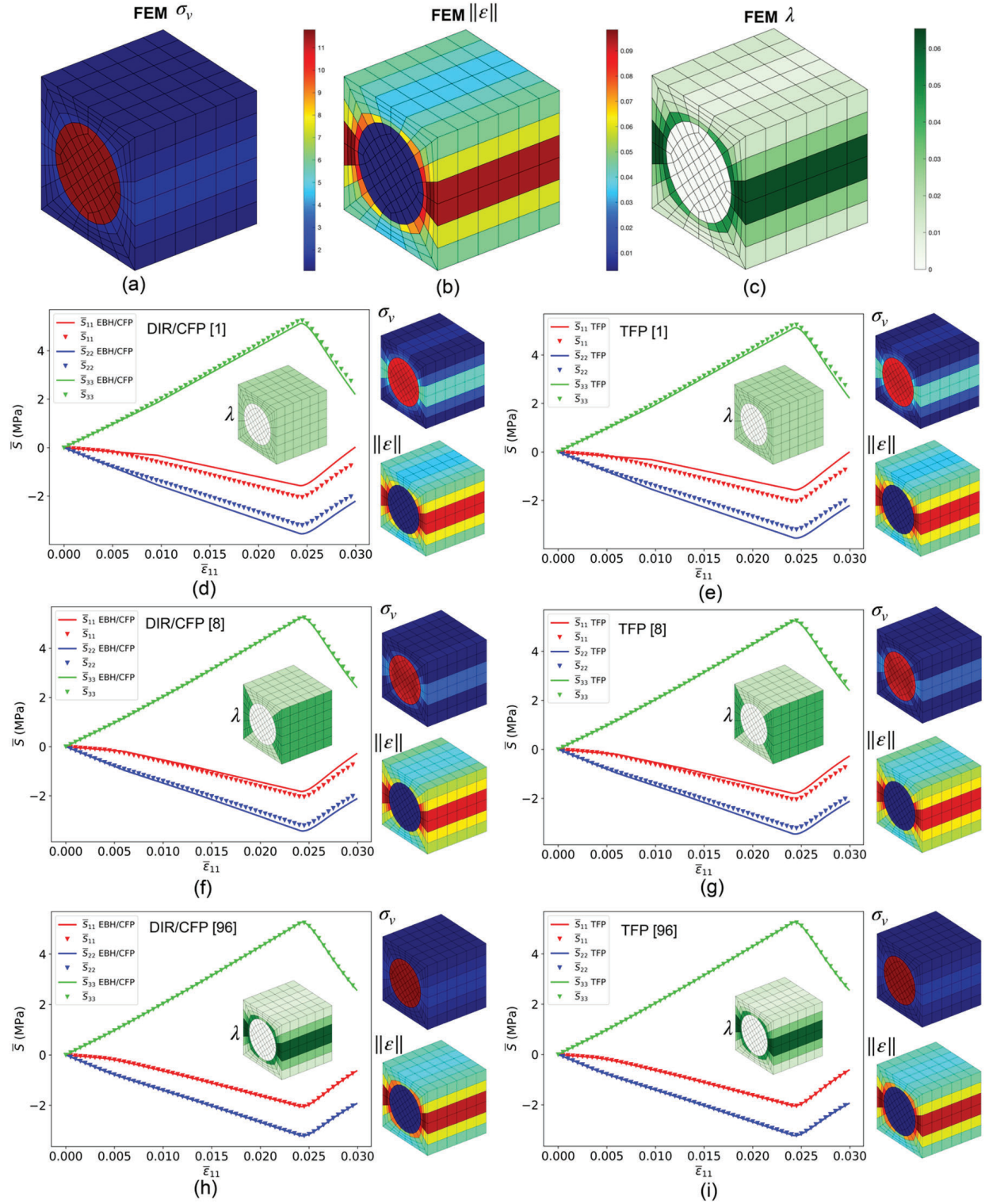


Fig. 6. EBH results for matrix modeled as elastoplastic under mixed-mode tension. (a)-(c) FEM solutions of fine-scale von Mises stress σ_v , strain norm $\|\epsilon\|$, and accumulated plastic multiplier λ , at the end of loading. (d), (f), and (h) show the DIR/CFP solutions for coarse-scale stress and the corresponding fine-scale fields computed with 1, 8, 96 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities of interest and partitions.

3.2.2.2. CFP convergence and the TFP variant. The partition level strain at an iteration $m + 1$, ${}^{m+1}\epsilon_{ij}^B$, and its converged counterpart $\hat{\epsilon}_{ij}^B$ are given as

$${}^{m+1}\epsilon_{ij}^B = \Sigma_{ij}^{klB} \bar{\epsilon}_{kl} + P_{ij}^{klBA} \left[\frac{\tau}{\tau + \Delta\gamma} q\mu_{kl}^A + \frac{\Delta\gamma}{\tau + \Delta\gamma} {}^m\omega^A {}^m\epsilon_{kl}^A \right] \quad (46)$$

$$\hat{\epsilon}_{ij}^B = \Sigma_{ij}^{klB} \bar{\epsilon}_{kl} + P_{ij}^{klBA} \left[\frac{\tau}{\tau + \Delta\gamma} q\mu_{kl}^A + \frac{\Delta\gamma}{\tau + \Delta\gamma} \hat{\omega}^A \hat{\epsilon}_{kl}^A \right] \quad (47)$$

where $\hat{\omega}^A$ is a converged damage variable. Subtracting Eq. (46) from (47) yields the partition strain error at $(m + 1)$ -th iteration, ${}^{m+1}e_{ij}^B$, given by

$${}^{m+1}e_{ij}^B = P_{ij}^{klBA} \frac{\Delta\gamma}{\tau + \Delta\gamma} [\hat{\omega}^A \hat{\epsilon}_{kl}^A - {}^m\omega^A {}^m\epsilon_{kl}^A] \quad (48)$$

Taking Euclidean norm of Eq. (48) and bounding

$$\| \hat{\omega}^A \hat{\epsilon}_{kl}^A - {}^m\omega^A {}^m\epsilon_{kl}^A \|_2 \leq \omega_{\max} \| {}^m\epsilon_{kl}^A \|_2 \quad (49)$$

where the $\omega_{\max} \geq \max_A(\hat{\omega}^A, {}^m\omega^A)$ yields

$$\| {}^{m+1}e_{ij}^B \|_2 \leq \frac{\omega_{\max} \Delta\gamma}{\tau + \Delta\gamma} \| P_{ij}^{klBA} \|_2 \| {}^m\epsilon_{kl}^A \|_2 \quad (50)$$

This completes the proof of monotonic error decrease with a rate of convergence governed by the increment of load parameter $\Delta\gamma$, viscous regularization parameter τ , and the largest partition damage parameter $\omega_{\max}$. It is evident that the CFP does not converge in absence of viscous stabilization $\tau = 0$ and $\omega_{\max} = 1$, which not surprisingly is shared with the classical Newton method.

Similarly to the plasticity case, we will study the performance of the truncated variant of the fixed-point iteration algorithm (TFP) in Section 4.

3.3. The EBH computational framework

Fig. 3 illustrates the workflow of the EBH computational framework for modeling inelastic behavior of nonlinear periodic materials, consisting of offline and online stages.

The goal of the offline stage is to compute the influence functions required for the online stage. The subdomain is subdivided into a set of partitions, each containing either a single finite element or a collection of elements. Since EBH treats eigenstates as the primary unknowns, the resulting system has naturally fewer degrees-of-freedom compared to a classical FEM. The influence functions are obtained by solving a sequence of elastic RVE problems, each subjected to coarse-scale strain as well as partition-level eigenstrains.

In the online stage, for each increment of the coarse-scale strain, the algorithm determines which partitions undergo inelastic processes in order to identify the active unknowns and assemble the corresponding system of nonlinear equations based on the local constitutive models. The updated eigenstates are then used in the post-processing step to compute the coarse- and fine-scale fields of interest. Once the residual converges, the procedure advances to the next coarse-scale strain increment.

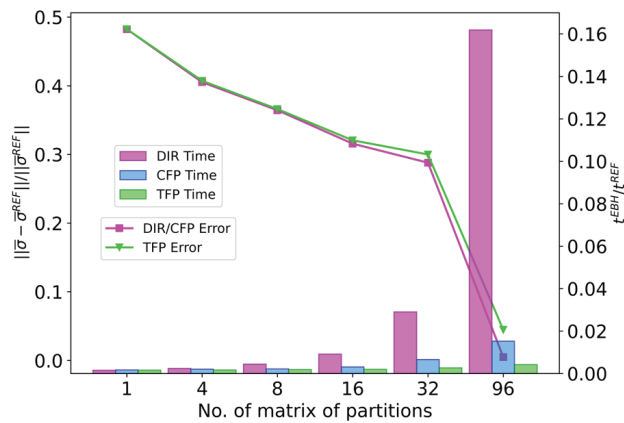


Fig. 7. Convergence of coarse-scale stress relative to FEM stress and normalized CPU time for the matrix modeled as elastoplastic for different partitions and EBH schemes.

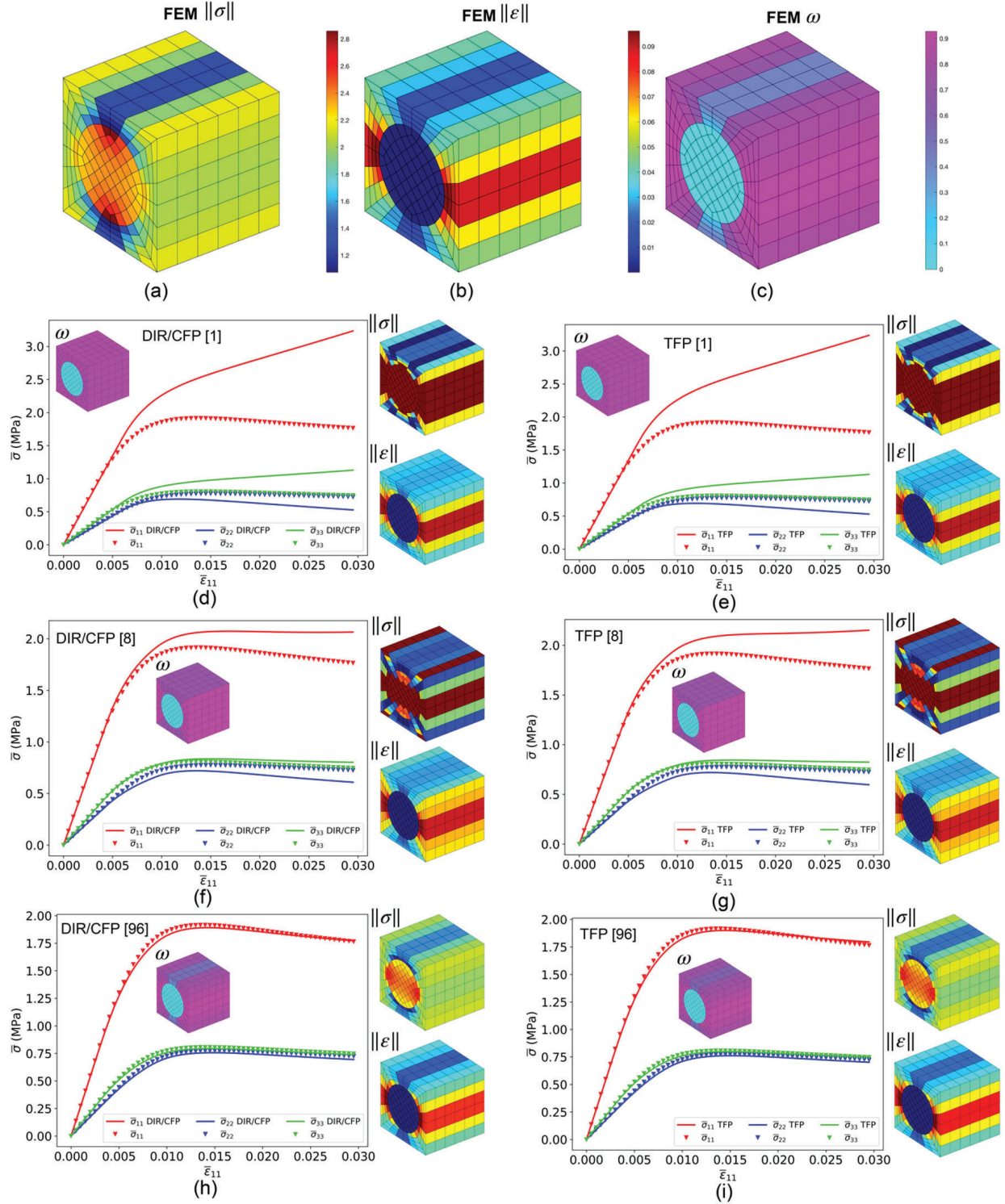


Fig. 8. EBH results for a matrix modeled by the regularized damage with coarse-scale softening under uniaxial tension applied normal to the fiber direction. (a)–(c) are the FEM solution of fine-scale stress norm $\|\sigma\|$, strain norm $\|\epsilon\|$, and damage variable ω at the end of loading. (d), (f), and (h) show the DIR/CFP solutions for coarse-scale stress and corresponding fine-scale fields computed with 1, 8, and 96 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities of interest and partition levels.

4. Numerical examples

This section demonstrates the workflow, accuracy, and computational efficiency of various EBH framework on single-fiber and randomly distributed four-fiber RVE models for both plasticity and regularized damage constitutive models. We consider three EBH solution schemes, namely direct solution of the coupled system of equations (DIR), as well as fixed-point iterative solutions, CFP and

TFP. The results are compared to FEM in terms of accuracy and computational efficiency for multiple load cases and various partition configuration.

4.1. Single-fiber RVE model

4.1.1. Geometry, material properties and discretization

Fig. 4(a) shows the geometry of the single-fiber RVE, where a circular cross-section fiber is centered in a unit cubic box representing composite material reinforced with uniformly distributed continuous fibers. The RVE is periodic along all three spatial directions. The side length is $L = 1\text{ mm}$, and the fiber radius is $r = 0.35\text{ mm}$, corresponding to a fiber volume fraction of approximately 40%. All numerical examples presented herein follow the coordinate system defined in Fig. 4(a). The finite element mesh, shown in Fig. 4(b), consists of 936 eight-node hexahedral elements, resulting in a total of 3549 DOFs for the FEM. The Young's modulus of the fiber and matrix are $E_1 = 10\text{ GPa}$ and $E_2 = 100\text{ MPa}$, respectively, and both phases share a Poisson's ratio of $\nu = 0.3$. The matrix is modeled using either elastoplastic or damage constitutive laws, while fibers are assumed to exhibit damage behavior.

The matrix subdomain is discretized into a series of partitions, ranging from the extremely coarse case—where the entire matrix is treated as a single partition—to the extremely fine case, where each strip of finite elements aligned with the fiber direction is defined as an individual partition. Fig. 4(c)–(f) illustrate the discretization of the matrix subdomain into 1, 4, 8, and 96 partitions, respectively, with the latter representing the finest configuration. The cases with 4 and 8 partitions are of specific interest since the numbers of DOFs is comparable to that obtained using one-partition per-phase approach [20–24].

4.1.2. Elastoplastic matrix and regularized damage model for fibers

In this series of examples, the matrix phase is modeled as elastoplastic with linear isotropic hardening, characterized by an initial yield stress of $Y_0 = 1\text{ MPa}$ and a plastic modulus of $H = 10\text{ MPa}$. The regularized damage behavior of the fiber is defined by the following parameters: $\varepsilon_0 = 0.0025$, $\varepsilon_f = 0.0032$ and $\tau = 10\Delta\gamma$.

Two load cases are considered: 1) Uniaxial tension applied normal to the fiber direction of up to $\bar{\varepsilon}_{11} = 0.03$; and 2) Mixed-mode tension applied along three principal directions of up to $\bar{\varepsilon}_{11} = 0.03$, $\bar{\varepsilon}_{22} = 0.018$ and $\bar{\varepsilon}_{33} = 0.001$.

Each load case is simulated using 60 increments. For every load case and partition configuration, we consider three EBH solution schemes, direct solution of the coupled system of equations (DIR), as well as fixed point solutions, CFP and TFP. The results are compared to FEM in terms of accuracy and computational efficiency. Both CFP and DIR are simulated to verify that CFP converges to DIR.

4.1.2.1. *Uniaxial tension normal to fiber.* Fig. 5 compares the results obtained from FEM and EBH solutions. First, we verify that the results obtained from DIR and CFP schemes are identical. Comparing the EBH schemes to the FEM solution, noticeable discrepancies are evident when the entire matrix is modeled as a single partition (Fig. 5d). Not surprisingly the underlying assumption of uniform plastic multiplier fails to capture the fine-scale behavior observed in the FEM solution. Perhaps a bit less surprising is its inability to capture the coarse-scale behavior (unless perhaps material properties are calibrated). Although the EBH results with eight partitions (Fig. 5f) show marginal improvement, rather considerable deviations from the reference fine-scale and coarse-scale solutions remain evident. In contrast, the EBH solution with 96 partitions (Fig. 5h) achieves nearly identical results to the FEM in both coarse-scale stress response and fine-scale fields, while maintaining substantially lowered computational cost, as will be illustrated in SubSection 4.1.2.3.

Interestingly, the differences between the results obtained using the TFP and those from CFP/DIR are negligible across all partition configurations, confirming the robustness and high accuracy of the TFP approach, particularly for the 96-partition case (Fig. 5i).

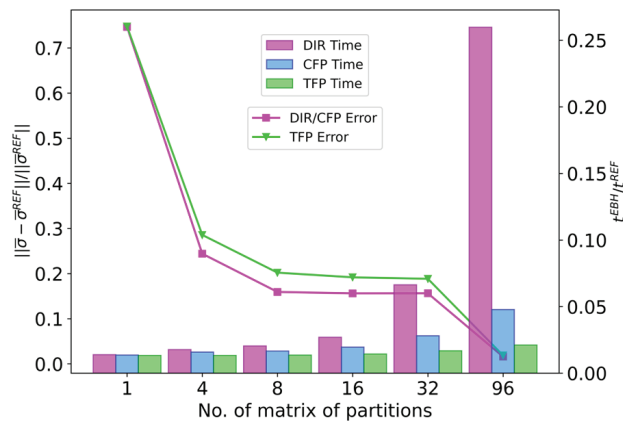


Fig. 9. Convergence of normalized coarse-scale stress with respect to the FEM solution and normalized CPU time for the case of matrix modeled by regularized damage model with coarse-scale softening for increasing levels and EBH schemes.

4.1.2.2. *Mixed-mode loading: tension along 1-, 2-, 3- axes.* Fig. 6 compares the FEM and EBH results under mixed-mode tension along the 1-, 2-, and 3- directions. Results suggest similar trend observed in the uniaxial tension case. The CFP solution with the finest partitioning is again in nearly perfect agreement with the FEM results (Fig. 6h).

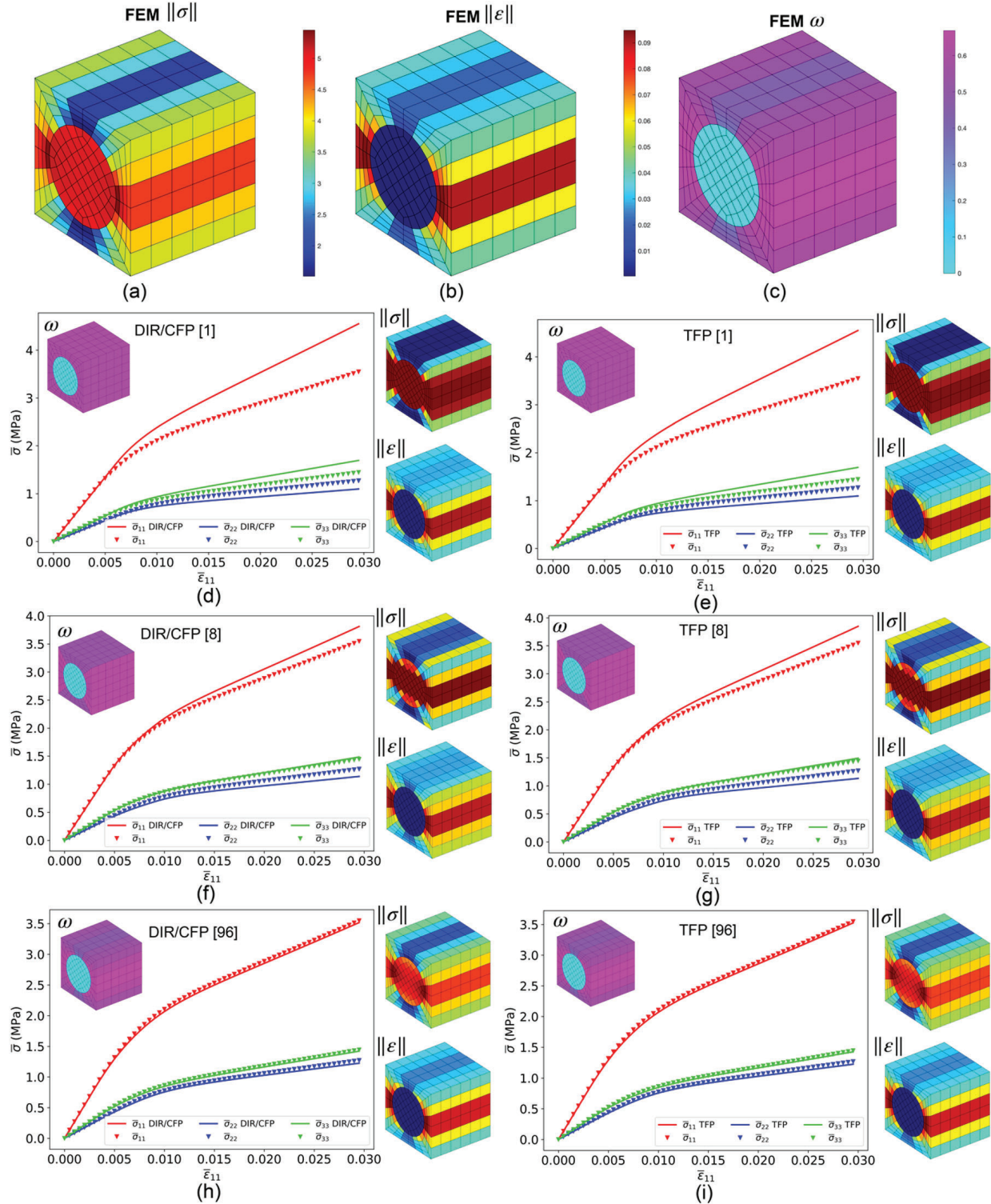


Fig. 10. EBH results for matrix modeled with regularized damage with coarse-scale hardening under uniaxial tension applied normal to the fiber direction. (a)-(c) FEM solutions of fine-scale stress norm $\|\sigma\|$, strain norm $\|\epsilon\|$, and damage variable ω , at the end of loading. (d), (f), and (h) show the DIR/CFP solutions for coarse-scale stress and corresponding fine-scale fields computed with 1, 8, and 96 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities and partition levels.

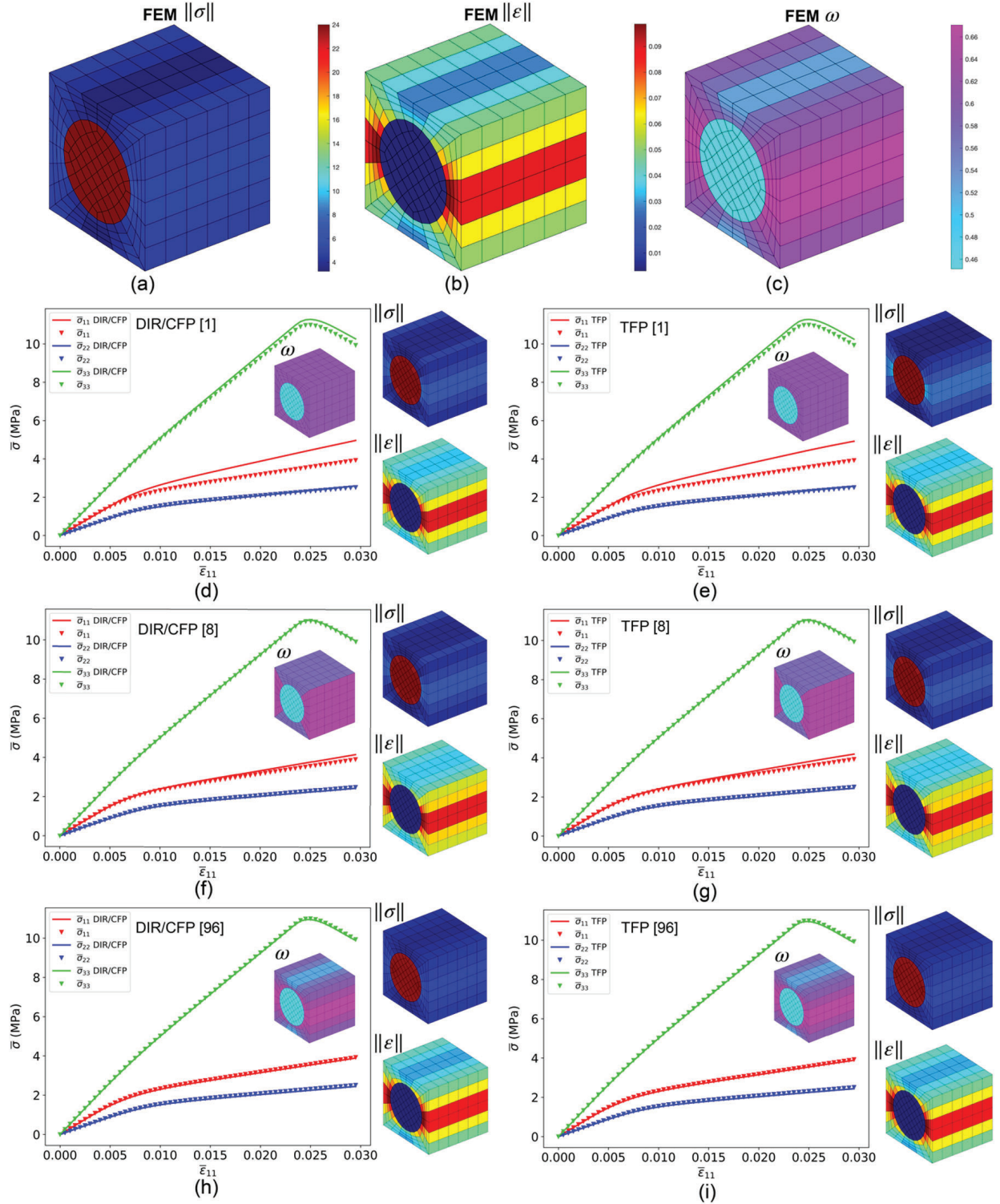


Fig. 11. EBH results for matrix modeled by regularized damage with coarse-scale hardening under mixed mode load. (a)-(c) FEM solutions of fine-scale stress norm $\|\sigma\|$, strain norm $\|\epsilon\|$, and damage variable ω at the end of loading. (d), (f), and (h) show the DIR/CFP solutions for coarse-scale stress and corresponding fine-scale fields computed with 1, 8, 96 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities and partition levels.

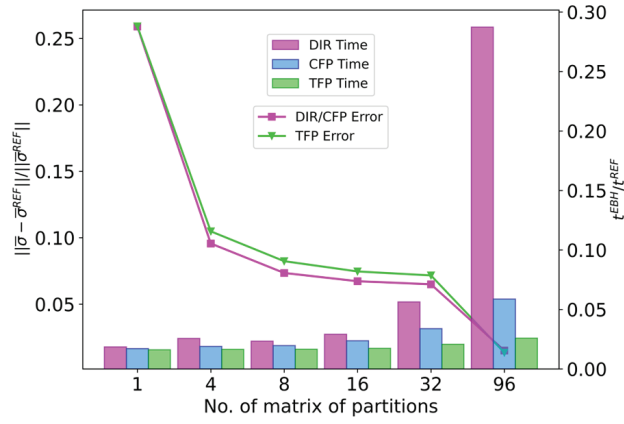


Fig. 12. Convergence of coarse-scale stress and normalized CPU time for the matrix modeled as regularized damage with coarse-scale hardening for different numbers of partitions and EBH schemes.

Table 1

Comparison of EBH and FEM condition numbers for plasticity and damage models.

Model/Method	FEM	DIR (96)	DIR (32)	DIR (16)	DIR (8)	DIR (4)	DIR (1)
Plasticity	2.87E+5	10.8	5.0	4.0	3.3	1.9	1
Damage	3.11E+5	250	10.3	8.9	6.4	12.8	1

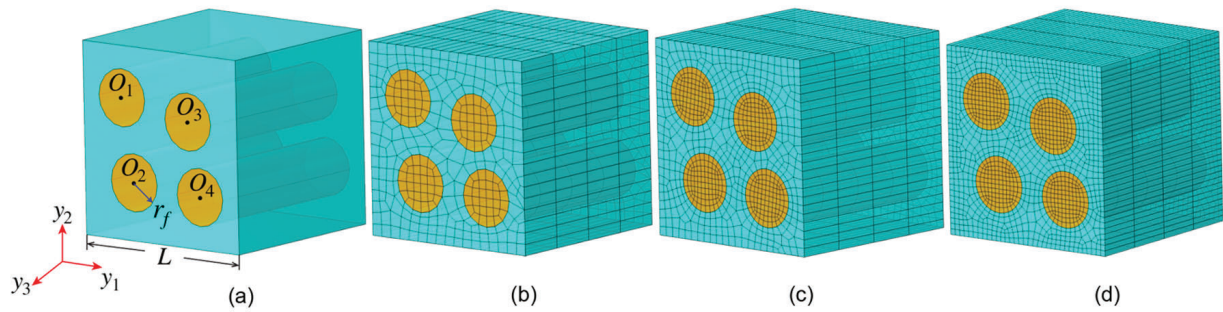


Fig. 13. Four-fiber RVE domain. (a) Geometry of the RVE. (b-d) Finite element meshes containing 1392, 3392, and 5036 eight-node hexahedral elements, respectively.

In this loading condition, the applied tension along the fiber direction induces damage within the fiber phase, reflected by the reduction in the coarse-scale deviatoric stress component s_{33} . The corresponding eigenstates include both the plastic multiplier λ within each matrix partition and the damage variable ω in the fiber, modeled as a single partition.

4.1.2.3. CPU time and coarse-scale stress error convergence. Fig. 7 illustrates the convergence of the normalized coarse-scale deviatoric stress as the number of partitions is increased. The error decreases monotonically with finer partitioning, with the TFP method exhibiting slightly higher errors than the fully converged CFP. The normalized CPU time, measured relative to the FEM, increases with the number of partitions. Among the three EBH solution schemes, the DIR requires the longest computational time due to repeated solution of large dense system of equations, while the TFP is the highest computational efficiency. A few noteworthy observations: (i) the CPU time of the CFP with 96 partitions is roughly about 1.5% of that required by the FEM, while achieving nearly identical coarse-scale stress predictions, (ii) the CPU time of the TFP with 96 partitions is nearly identical to that of DIR with 8 partitions. The latter observation is meaningful since the number of unknowns in the DIR with 8 partitions is comparable to that of broadly-practiced one-partition per phase ROH approach [20–24].

4.1.3. Matrix modeled by regularized damage model with coarse-scale softening

In this series of examples, the matrix is modeled using a regularized damage constitutive law characterized by the following parameters: $\varepsilon_0 = 0.01$, $\varepsilon_f = 0.1$, $\tau = 10\Delta\gamma$ and $\omega_m = 0.97$. A uniaxial tension load is applied normal to the fiber direction up to $\bar{\varepsilon}_{11} = 0.03$, using 60 increments. For each domain partitioning configuration, all three EBH solution schemes, DIR, CFP and TFP, are compared in terms of accuracy and computational efficiency.

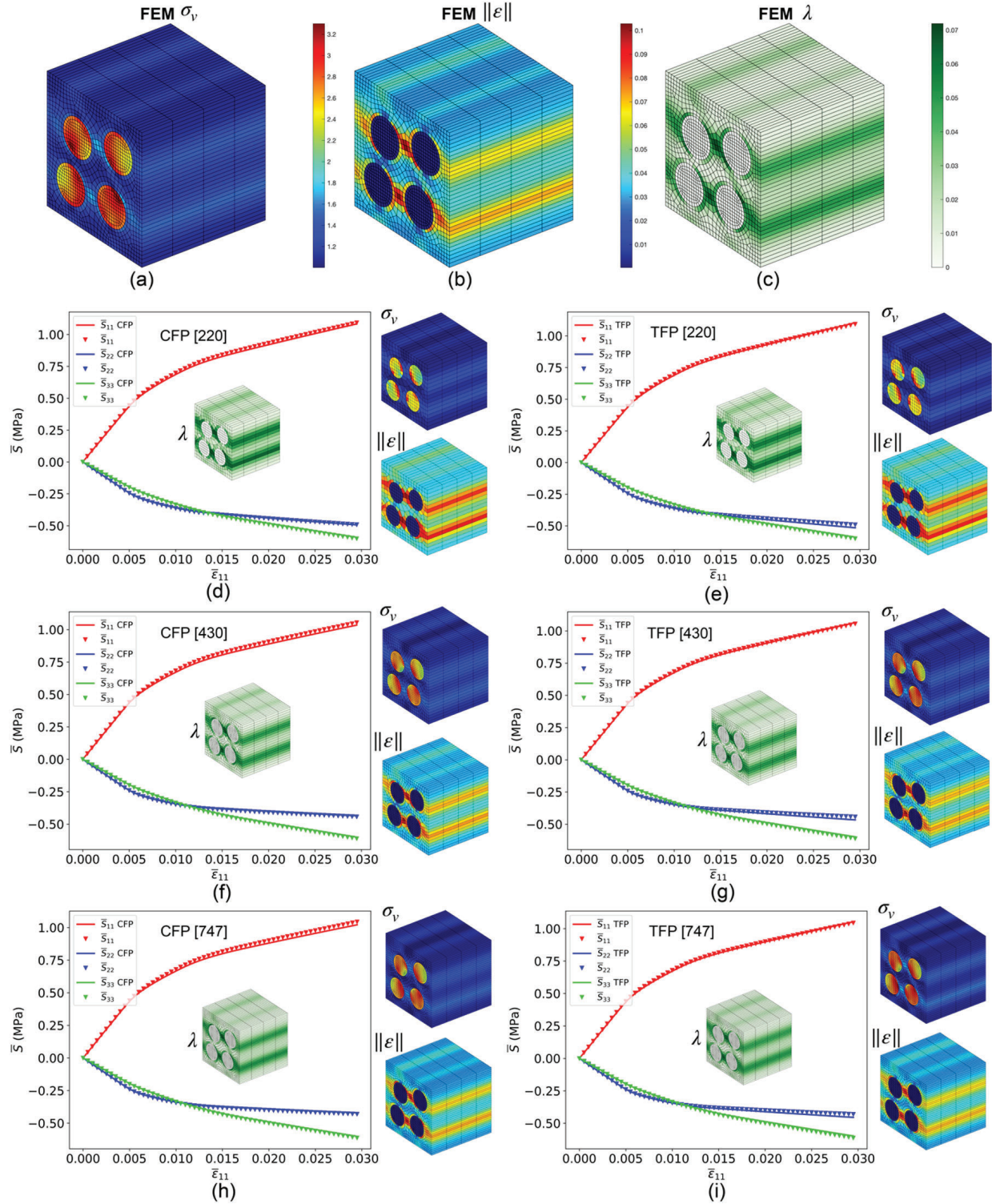


Fig. 14. EBH results for matrix modeled as elastoplastic. (a)-(c) FEM solutions of fine-scale von Mises stress σ_v , strain norm $\|\epsilon\|$, and accumulated plastic multiplier λ , computed with the finest mesh at the end of loading. (d), (f), and (h) show the CFP solutions for coarse-scale stress and corresponding fine-scale fields computed with 220, 430 and 747 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities and partition levels.

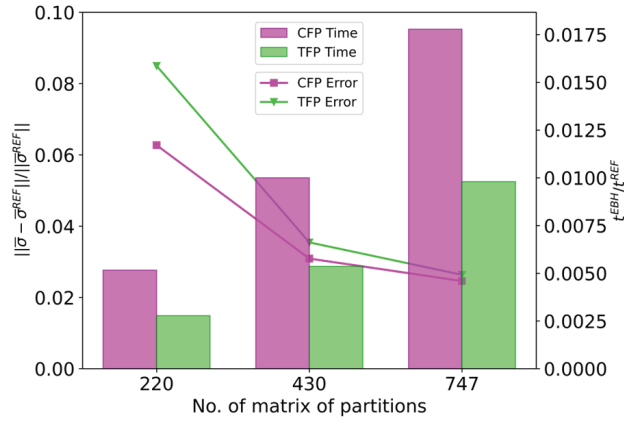


Fig. 15. Convergence of coarse-scale stress relative to the FEM solution on the finest mesh and normalized CPU time for a matrix modeled as elastoplastic for different partition numbers and two EBH schemes, CFP and TFP.

4.1.3.1. Uniaxial tension normal to fiber. Fig. 8 compares the FEM and EBH solutions for the matrix modeled with regularized damage constitutive law with coarse-scale softening. The general trend is consistent with the elastoplastic case. A coarse-scale softening behavior is observed at higher strain levels, $\bar{\epsilon}_{11} > 0.01$. The CFP solution of the coarse-scale stress with the finest partitioning is in excellent agreement with the FEM results (Fig. 8h). However, the corresponding fine-scale stress distribution is reasonably accurate but not perfect. This deviation primarily arises due to non uniqueness of the fine-scale solution even with viscous regularization, and consequently, to the sensitivity of the solution to the mesh size and the discretization scheme (FEM vs EBH) in general.

4.1.3.2. CPU time and stress error convergence. Fig. 9 illustrates the convergence of the coarse-scale stress with increasing numbers of partitions. Two noteworthy observations: (i) the CPU time of the CFP with 96 partitions is roughly 5% of the FEM simulation, while achieving nearly identical coarse-scale stress predictions, (ii) the CPU time of the TFP approach with 96 partitions is again nearly identical to that of DIR with 8 partitions.

4.1.4. Matrix modeled as regularized damage with coarse-scale hardening

In this series of examples, the matrix is modeled with regularized damage constitutive law characterized by the following parameters: $\epsilon_0 = 0.01$, $\epsilon_f = 0.1$, $\tau = 10\Delta\gamma$ and $\omega_m = 0.7$, selected to prevent coarse-scale stress softening. Both uniaxial and mixed-mode load cases described in Section 4.2 are studied.

4.1.4.1. Uniaxial tension normal to fiber. Fig. 10 compares the results obtained from EBH and FEM for matrix modeled with regularized damage with coarse-scale hardening under uniaxial tension applied normal to the fiber direction. In contrast to the coarse-scale softening case discussed in Subsection 4.3, the fine-scale stress distribution is accurately reproduced by both the CFP and TFP approaches with 96 partitions.

4.1.4.2. Mixed-mode loading: tension along 1-, 2-, 3- axes. Fig. 11 illustrates the EBH and FEM results for the mixed-mode loading case. It can be seen that both the CFP and TFP with 96 partitions are in good agreement with the FEM solution for both the coarse- and fine-scale quantities of interest.

4.1.4.3. CPU time and stress error convergence. As is shown in Fig. 12, the trends of both convergence in the normalized coarse-scale stress and normalized CPU time are consistent with those observed previously. The CPU time of the CFP with 96 partitions is only 5% of FEM, further demonstrating high computational efficiency and accuracy of CFP and TFP.

4.1.5. Comparison of condition numbers

In this section we compare the condition number of the tangent stiffness matrix in the direct EBH method or DIR with that of the FEM. A low condition of the tangent stiffness matrix could offer a possibility in employing an iterative solver within the Newton method for solving a dense linear system of equations.

Furthermore, a low condition number of the tangent stiffness matrix bodes well with any iterative approach and is another indication for very fast convergence of the proposed CFP and TFP approaches. Table 1 depicts condition numbers for DIR(x) with x number of partitions in comparison FEM for plasticity and regularized damage constitutive models. With 96 partitions the EBH DIR condition number is three to four orders of magnitude lower than that of FEM.

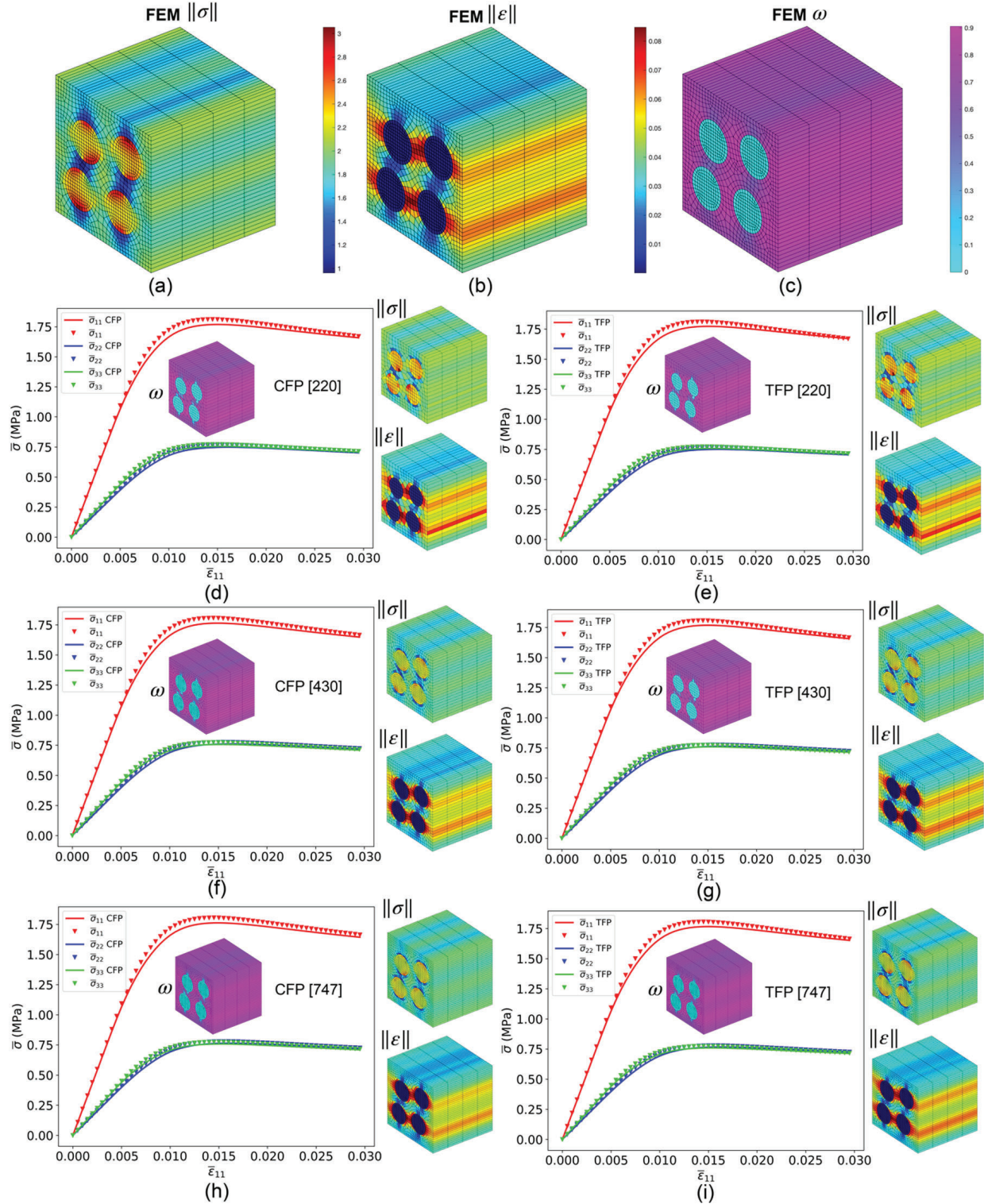


Fig. 16. EBH results for matrix modeled as regularized damage with coarse-scale softening. (a-c) FEM solution of fine-scale stress norm $\|\sigma\|$, strain norm $\|\epsilon\|$, and damage variable ω , computed with the finest mesh at the end of loading. (d), (f), and (h) show the CFP solutions for coarse-scale stress and corresponding fine-scale fields computed with 220, 430 and 747 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities and partition levels.

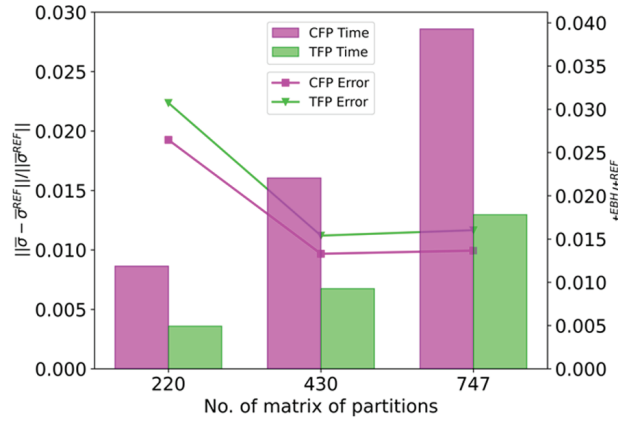


Fig. 17. Convergence of coarse-scale stress to the FEM solution on the finest mesh and normalized CPU time for the matrix modeled as regularized damage with coarse-scale softening for threepartition levels and CFP and TFP schemes.

4.2. Four-fiber RVE model

Herein we study EBH performance on four-fiber RVE problem. Results are compared to the finest finite element mesh for both plasticity and regularized damage constitutive models.

4.2.1. Geometry and discretization

Fig. 13(a) illustrates the geometry of an RVE containing four randomly distributed fibers. The RVE is periodic along all three spatial directions. The geometric parameters are as follows: side length of the cubic RVE, $L = 1 \text{ mm}$, and the radius of each fiber $r = 0.15 \text{ mm}$ corresponding to a fiber volume fraction of approximately 30%. The fiber centers are located at: O_1 : ($y_1 = -0.25, y_2 = 0.23$), O_2 : ($y_1 = -0.18, y_2 = -0.2$), O_3 : ($y_1 = 0.18, y_2 = 0.15$), and O_4 : ($y_1 = 0.25, y_2 = -0.23$) with the origin located at the center of the RVE.

Fig. 13(b-d) illustrates the three mesh levels, consisting of 1392, 3392, and 5036 eight-node hexahedral elements, respectively. The material properties, maximum loads, and loading rates are identical to those used in the corresponding cases in Section 4. Only uniaxial tension applied normal to the fiber direction is considered in this section. The EBH analyses employ the finest partitioning scheme for each mesh configuration, yielding 220, 430, and 747 partitions in the matrix, while each fiber is modeled as a single partition. Due to the large number of partitions, only the CFP and TFP methods are studied. All EBH results are compared against FEM results on the finest mesh (Fig. 13d).

4.2.2. Matrix modeled as elastoplastic and fibers as regularized damage

Fig. 14 compares the EBH and FEM results for the case with an elastoplastic matrix. All partition configurations yield sufficiently accurate coarse-scale stress responses. However, the fine-scale stress field predicted with the coarsest partitioning (i.e., 220 partitions; Fig. 14d) shows slight deviations from the FEM solution. These discrepancies diminish with finer partitions, and the results obtained with finest partition (Fig. 14h-i) closely match the FEM predictions.

Fig. 15 illustrates the convergence of the coarse-scale stress to the FEM solution and the corresponding normalized CPU time. All stress errors remain relatively small and decrease consistently as the number of partitions increases. The normalized CPU time increases approximately linearly with the number of partitions. Notably, the CPU time for the finest partition configuration is only about 1% of that required by the FEM simulation, highlighting the exceptional computational efficiency of the EBH framework.

4.2.3. Matrix modeled as regularized damage with coarse-scale softening

Fig. 16 compares the EBH and FEM results for the case of matrix modeled with regularized damage constitutive law exhibiting coarse-scale softening. The trends observed in both the coarse-scale and fine-scale responses are consistent with that discussed in SubSection 4.1.2. The fine-scale stress is predicted reasonably well for all three partition levels, indicating that the finer EBH partitions can reproduce an eigenstrain field sufficiently close to the reference solution.

Fig. 17 illustrates the convergence of the coarse-scale stress and normalized CPU time both of which are consistent with the observations discussed in SubSection 4.1.2.

4.2.4. Matrix modeled as damage with coarse-scale hardening

Fig. 18 compares the EBH and FEM results for the case of the matrix modeled with regularized damage constitutive law exhibiting coarse-scale hardening. The trends observed in both the coarse-scale and fine-scale responses are consistent with those discussed in SubSection 4.1.2. The fine scale stress is predicted with sufficient accuracy across all partition levels. Fig. 19 illustrates the convergence of the coarse-scale stress and the corresponding normalized CPU time, both of which are consistent with the observations reported earlier.

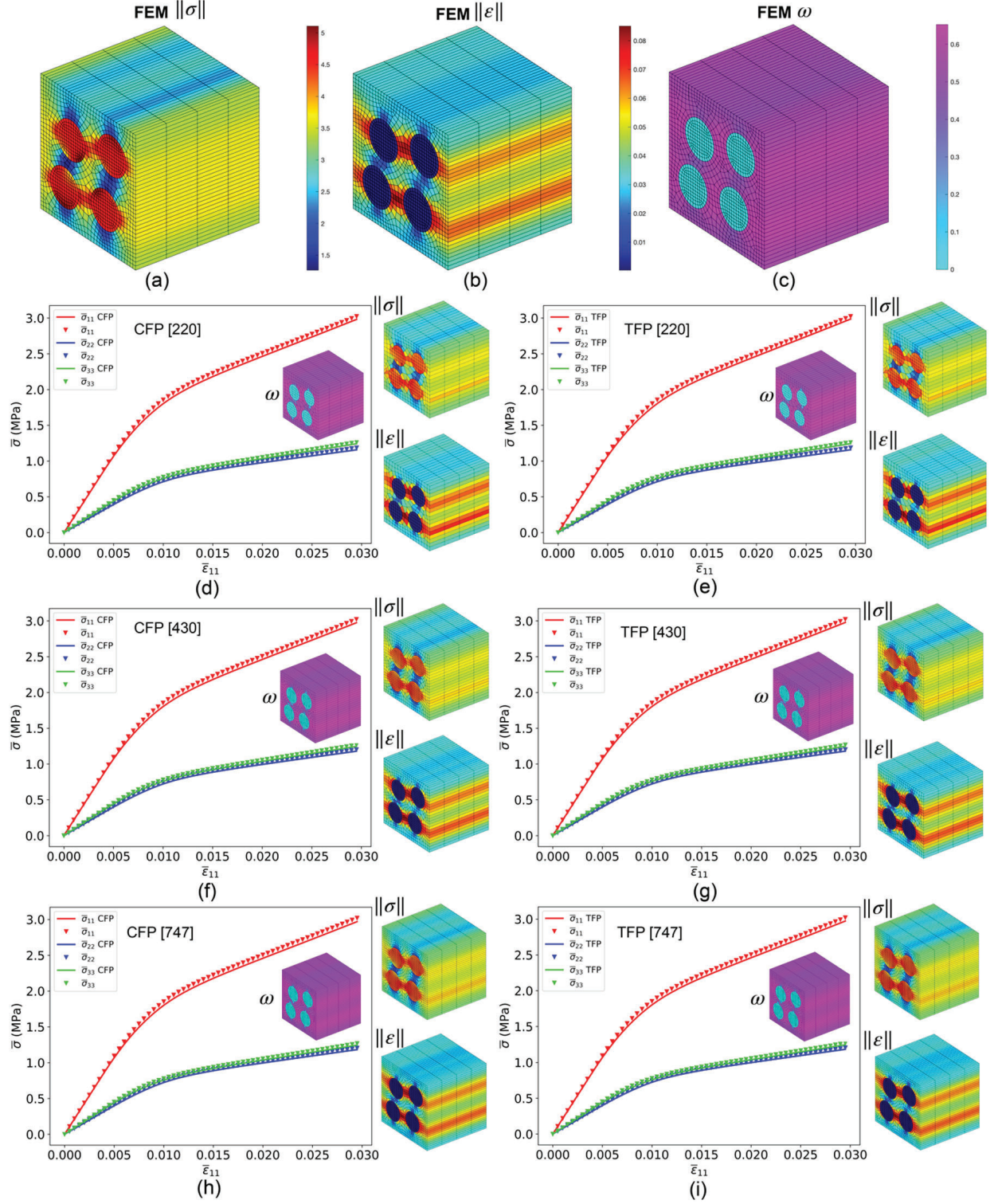


Fig. 18. EBH results for matrix modeled as regularized damage with coarse-scale hardening. (a)-(c) FEM solutions of fine-scale stress norm $\|\sigma\|$, strain norm $\|\epsilon\|$, and damage variable ω , computed with the finest mesh at the end of the load. (d), (f), and (h) show the CFP solutions for coarse-scale stress and corresponding fine-scale fields computed with 220, 430 and 747 partitions, respectively. (e), (g), and (i) present the TFP solutions for the same quantities and partition levels.

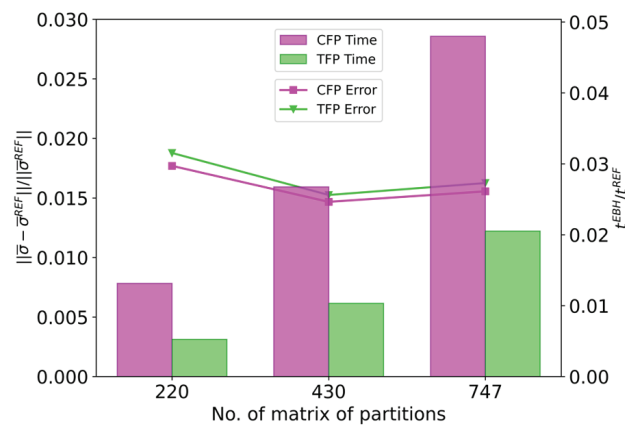


Fig. 19. Convergence of coarse-scale stress to the FEM solution on the finest mesh and normalized CPU time for the matrix modeled as regularized damage with coarse-scale hardening for three partitions and CFP and TFP schemes.

5. Conclusions and future work

This work introduced the Eigenstate-Based Homogenization (EBH) framework, a new homogenization strategy designed to combine the accuracy of fully resolved computational homogenization with the efficiency traditionally associated with unresolved or reduced-order homogenization methods [20–24]. The EBH method belongs to the broader class of transformation field analysis approaches but incorporates two key innovations: an eigenstate-based representation of the fine-scale solution space and a Controllable Fixed-Point (CFP) solver, together with its truncated variant, for the nonlinear evolution of state variables.

The numerical studies conducted on single-fiber and four-fiber RVEs confirm that EBH provides coarse-scale responses in excellent agreement with fully resolved FEM simulations while offering orders-of-magnitude reductions in computational cost. Fine-scale fields are also captured with high fidelity when the matrix domain is partitioned at sufficiently fine resolution. For both plasticity and regularized damage models, the CFP method demonstrated monotonic convergence, with rates governed primarily by the load parameter increment, viscous regularization and maximum damage variable (for damage), and the magnitude of the active eigenstate increment (for plasticity).

In addition to the fully converged CFP solver, a truncated variant (TFP) was examined. TFP exploits the rapid error reduction observed in the initial CFP iterations and achieves solutions of practical accuracy at a fraction of the computational cost. Across all examples, the CPU time of the TFP method was found to be comparable to that of conventional reduced-order homogenization approaches employing one partition per phase, while yielding significantly improved accuracy at the coarse scale without constitutive model calibration, with the added benefit of retaining high accuracy at the fine scale.

Overall, the EBH framework provides an efficient and accurate alternative to direct homogenization methods for nonlinear multiscale analysis. The eigenstate representation sharply reduces the number of unknowns, while the CFP and TFP solvers provide robust and scalable algorithms suitable for large numbers of partitions.

Furthermore, this work establishes a critical methodological distinction between classical parameter calibration and the proposed "parameter readjustment" strategy. By utilizing manufacturer specifications as a high-fidelity prior and restricting the optimization process to physical batch variations, the proposed method effectively decouples true material variability from numerical discretization errors. This separation is identified as a theoretical prerequisite for the advancement of probabilistic multiscale frameworks. Unlike traditional calibration, which risks conflating aleatory uncertainty (material randomness) with epistemic uncertainty (numerical error), the readjustment approach ensures that identified parameters remain physically consistent. This provides a rigorous foundation for future stochastic modeling where material variability must be treated as a distinct and uncontaminated random process.

Future work will focus on extending EBH to:

- consideration of more complex nonlinearities, such as kinematic hardening and anisotropic damage;
- adaptive partition refinement, enabling automatic control of accuracy versus cost;
- coupled multi-physics problems, particularly thermo-mechanical and chemo-mechanical systems; and
- consideration of different types of finite elements.

CRedit authorship contribution statement

Jacob Fish: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Junhe Cui:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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